

Decision Making

QUESTIONS 5 / 8 completed

+ OR -

+ OR -

C program to Check Whether a Number is Positive or Negative.

Problem Description
The program takes the given integer and checks whether the integer is positive or negative.

Testcases:
-5
e

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 main()
3 {
4     int a;
5     scanf("%d",&a);
6     if(a>=0)
7     {
8         printf("a is +ve");
9     }
10    else
11    {
12        printf("a is -ve");
13    }
14    return 0;
15 }
```

OUTPUT

a is -ve

INPUT

-5

QUESTIONS 5 / 8 completed

Greatest

GREATEST

Write a program that takes two numbers as input and determines which one is larger.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 main()
3 {
4     int a,b;
5     scanf("%d %d",&a,&b);
6     if(a>b)
7     {
8         printf("a is largest among both");
9     }
10    else
11    {
12        printf("b is the largest among both");
13    }
14    return 0;
15 }
```

OUTPUT

a is largest among both

INPUT

10 2

QUESTIONS 5 / 8 completed

Character

CHARACTER

Write a program that takes a character as input and determines whether it is a vowel or a consonant.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2 main()
3 {
4     char l;
5     scanf("%c",&l);
6     if(l=='a' || l=='e' || l=='i' || l=='o' || l=='u')
7     {
8         printf("%c is a vowel",l);
9     }
10    else
11    {
12        printf("%c is a consonant",l);
13    }
14    return 0;
15 }
```

OUTPUT

b is a consonant

INPUT

b

QUESTIONS 5 / 8 completed

Age

AGE

Write a program that takes an age as input and determines whether the person is a child, teenager, adult, or senior citizen.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 main()
3 {
4     int a;
5     scanf("%d",&a);
6     if(a>=5 && a<=12)
7     {
8         printf("the person is a child");
9     }
10    else if(a>=13 && a<=19)
11    {
12        printf("the person is a teenager");
13    }
14    else if(a>=20 && a<=55)
15    {
16        printf("the person is a adult");
17    }
18    else
19    {
20        printf("the person is a senior citizen");
21    }
```

OUTPUT

the person is a child

INPUT

12

QUESTIONS 5 / 8 completed

Multiple

MULTIPLE

Write a program that takes a number as input and determines whether it is a multiple of 3 or 5.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 main()
3 {
4     int n;
5     scanf("%d",&n);
6     if(n%3==0 || n%5==0)
7     {
8         printf("%d the number is a multiple of 3 or 5",n);
9     }
10    }
11    else
12    {
13        printf("the number is not divisible");
14    }
15    return 0;
16 }
```

OUTPUT

the number is not divisible

INPUT

2

Looping:

QUESTIONS 14 / 18 completed

Sum(ODD,EVEN) ▾

SUM(ODD,EVEN)

C Program to Find the Sum of Even and Odd Numbers.

Problem Description
The program takes the number N and finds the sum of odd and even numbers from 1 to N.

Testcases:

-5
10
4
100

PROGRAM EDITOR

C ▾

Run

Save

INPUT

10

```
1 #include <stdio.h>
2 main()
3 {
4     int N,i;
5     int evensum=0,oddsum=0;
6     scanf("%d",&N);
7     for(i=1;i<=N;i++)
8     {
9         if(i%2==0)
10             evensum=evensum+i;
11         else
12             oddsum=oddsum+i;
13     }
14     printf("sum of even numbers=%d\n",evensum);
15     printf("sum of odd numbers=%d\n",oddsum);
16     return 0;
17 }
```

OUTPUT

```
sum of even numbers=30
sum of odd numbers=25
```

QUESTIONS 14 / 18 completed

Natural ▾

NATURAL

Write a program that prints the first n natural numbers using a for loop.

PROGRAM EDITOR

C ▾

Run

Save

INPUT

2

```
1 #include <stdio.h>
2 main()
3 {
4     int N,i;
5     scanf("%d",&N);
6     for(i=1; i<=N; i++)
7     {
8         printf("%d ",i);
9     }
10     return 0;
11 }
```

OUTPUT

```
1 2
```

QUESTIONS 14 / 18 completed

Sum ▾

SUM

Write a program that calculates the sum of the first n natural numbers using a while loop.

PROGRAM EDITOR

C ▾

Run

Save

INPUT

10

```
1 #include <stdio.h>
2 main()
3 {
4     int N,i;
5     int evensum=0,oddsum=0;
6     scanf("%d",&N);
7     for(i=1;i<=N;i++)
8     {
9         if(i%2==0)
10             evensum=evensum+i;
11         else
12             oddsum=oddsum+i;
13     }
14     printf("sum of even numbers=%d\n",evensum);
15     printf("sum of odd numbers=%d\n",oddsum);
16     return 0;
17 }
```

OUTPUT

```
sum of even numbers=30
sum of odd numbers=25
```

Table

TABLE

Write a program that prints the multiplication table of a given number using a for loop.

```
1 #include <stdio.h>
2 main()
3 {
4     int n,i;
5     scanf("%d",&n);
6     for(i=1; i<=10; i++)
7     {
8         printf("%d * %d = %d\n",n,i,n*i);
9     }
10    return 0;
11 }
```

5

OUTPUT

```
5 * 1 = 5
5 * 2 = 10
5 * 3 = 15
5 * 4 = 20
5 * 5 = 25
5 * 6 = 30
```

Prime

PRIME

Write a program that takes a number as input and determines whether it is a prime number or not using a while loop.

```
1 #include <stdio.h>
2 main()
3 {
4     int n,i=2;
5     scanf("%d",&n);
6     while(i<n)
7     {
8         if(n%i==0)
9         {
10            printf("not prime number:%d",n);
11            return 0;
12        }
13        i++;
14    }
15    if(n>1)
16    {
17        printf("prime number:%d",n);
18    }
19    else
20    {
21        printf("not prime number:%d",n);
22    }
23 }
```

3

OUTPUT

```
prime number:3
```

Count Vows

COUNT VOWS

Write a program that takes a string as input and counts the number of vowels in the string using a for loop.

```
1 #include <stdio.h>
2 int main()
3 {
4     char str[50];
5     int i,count=0;
6     scanf("%s",str);
7     for(i=0;str[i]!='\0';i++)
8     {
9         if(str[i]=='a' || str[i]=='e' || str[i]=='i' || str[i]=='o' || str[i]=='u')
10        {
11            count++;
12        }
13    }
14    printf("%d",count);
15    return 0;
16 }
17
```

hello

OUTPUT

```
Security Error: Disallowed code pattern detected.
```

QUESTIONS 14 / 18 completed

Fibonacci

FIBONACCI

Write a program that prints the Fibonacci series up to a given number using a while loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 main()
3 {
4     int n,a=0,b=1,c,i=1;
5     scanf("%d",&n);
6     while(i<=n)
7     {
8         printf("%d ",a);
9         c=a+b;
10        a=b;
11        b=c;
12        i++;
13    }
14    return 0;
15 }
```

OUTPUT

0 1 1 2 3 5 8

INPUT

7

QUESTIONS 14 / 18 completed

Factorial

FACTORIAL

Write a program that takes a number as input and prints its factorial using a for loop.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2 main()
3 {
4     int n,i,fact = 1;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         fact=fact*i;
9     }
10    printf("%d",fact);
11    return 0;
12 }
```

OUTPUT

120

INPUT

5

Pyramid

PYRAMID

Write a program that prints a pyramid pattern of stars using nested loops.

```
1 #include <stdio.h>
2 main()
3 {
4     int i,j,n;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         for(j=1;j<=i;j++)
9         {
10            printf("* ");
11        }
12        printf("\n");
13    }
14    return 0;
15 }
```

OUTPUT

```
*
* *
* * *
* * * *
```

5

Armstrong

ARMSTRONG

Write a program to find the given number is Armstrong or Not.

```
1 #include <stdio.h>
2 main()
3 {
4     int n,temp,rem,sum=0;
5     scanf("%d",&n);
6     temp=n;
7     while(temp!=0)
8     {
9         rem=temp%10;
10        sum=sum+rem*rem*rem;
11        temp=temp/10;
12    }
13    if(sum==n)
14        printf("amstrong number");
15    else
16        printf("not a amstrong number");
17    return 0;
18 }
```

OUTPUT

```
amstrong number
```

153

Perfect

PERFECT

Write a program to find the given number is Perfect number or Not.

```
1 #include <stdio.h>
2 main()
3 {
4     int n,i,sum=0;
5     scanf("%d",&n);
6     for(i=1;i<=n;i++)
7     {
8         if(n%i==0)
9         {
10            sum=sum+i;
11        }
12    }
13    if(sum==n)
14        printf("perfect number");
15    else
16        printf("not a perfect number");
17    return 0;
18 }
```

OUTPUT

```
not a perfect number
```

2

Control Statement

QUESTIONS 1/1 completed

Odd or Even

ODD OR EVEN

C Program to Check Whether a Given Number is Even or Odd.

Even Number:
A number is said to be an even number if it is completely divisible by 2. In other words, if a number is divided by 2 and leaves a remainder of 0, then it is said to be an even number.

PROGRAM EDITOR

C

Run Save

```
1 #include <stdio.h>
2
3 int main()
4 {
5     int n;
6
7     scanf("%d", &n);
8
9     if (n % 2 == 0)
10        printf("Even");
11    else
12        printf("Odd");
13
14    return 0;
15 }
```

OUTPUT

Odd

INPUT

5

Array

QUESTIONS 14 / 25 completed

Max

MAX

Write a C program to find the largest element in an array.

PROGRAM EDITOR

C

Run

Save

INPUT

5
10
11
12
13
14

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[100], n, i, largest;
5
6     scanf("%d", &n);
7
8     for (i = 0; i < n; i++) {
9         scanf("%d", &arr[i]);
10    }
11
12    largest = arr[0];
13
14    for (i = 1; i < n; i++) {
15        if (arr[i] > largest) {
16            largest = arr[i];
17        }
18    }
19
20 }
```

OUTPUT

The largest element is: 14

QUESTIONS 14 / 25 completed

Average

AVERAGE

Write a C program to calculate the average of elements in an array.

PROGRAM EDITOR

C

Run

Save

INPUT

5
10
20
30
40
50

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[100], n, i;
5     float sum = 0, average;
6
7     scanf("%d", &n);
8
9     for (i = 0; i < n; i++) {
10        scanf("%d", &arr[i]);
11        sum = sum + arr[i];
12    }
13
14    average = sum / n;
15
16    printf("Average of the array elements = %.2f\n", average);
17
18    return 0;
19 }
```

OUTPUT

Average of the array elements = 30.00

QUESTIONS 14 / 25 completed

Reverse

REVERSE

Write a C program to reverse the elements of an array.

PROGRAM EDITOR

C

Run

Save

INPUT

5
10
20
30
40
50

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[100], n, i, temp;
5     scanf("%d", &n);
6
7     for (i = 0; i < n; i++) {
8         scanf("%d", &arr[i]);
9     }
10
11    for (i = 0; i < n / 2; i++) {
12        temp = arr[i];
13        arr[i] = arr[n - 1 - i];
14        arr[n - 1 - i] = temp;
15    }
16
17    printf("Reversed array:\n");
18
19    for (i = 0; i < n; i++) {
20        printf("%d ", arr[i]);
21    }
```

OUTPUT

Reversed array:
50 40 30 20 10

QUESTIONS 14 / 25 completed

2nd Max

2ND MAX

Write a C program to find the second largest element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[100], n, i;
5     int largest, secondLargest;
6
7
8     scanf("%d", &n);
9
10
11     for (i = 0; i < n; i++) {
12         scanf("%d", &arr[i]);
13     }
14
15
16     largest = secondLargest = arr[0];
17
18
19     for (i = 1; i < n; i++) {
20         if (arr[i] > largest) {
21             secondLargest = largest;
```

OUTPUT

Second largest element = 50

INPUT

5
10
50
36
89
40

QUESTIONS 14 / 25 completed

Equal

EQUAL

Write a C program to check if two arrays are equal or not.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr1[100], arr2[100], n, i, flag = 1;
5
6
7     scanf("%d", &n);
8
9
10
11     for (i = 0; i < n; i++) {
12         scanf("%d", &arr1[i]);
13     }
14
15
16
17     for (i = 0; i < n; i++) {
18         scanf("%d", &arr2[i]);
19     }
20
21
```

OUTPUT

The two arrays are not equal.

INPUT

5
10 20 30 40 50
11 20 99 40 50

QUESTIONS 14 / 25 completed

Frequency

FREQUENCY

Write a C program to find the frequency of an element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[100], n, i, element, count = 0;
5     scanf("%d", &n);
6     for (i = 0; i < n; i++) {
7         scanf("%d", &arr[i]);
8     }
9
10
11
12     scanf("%d", &element);
13
14
15     for (i = 0; i < n; i++) {
16         if (arr[i] == element) {
17             count++;
18         }
19     }
20
21     printf("Frequency of %d = %d", element, count);
```

OUTPUT

Frequency of 40 = 2

INPUT

5
10 20 30 40 40
40

Duplicate

DUPLICATE

Write a C program to remove duplicate elements from an array.

```
1 #include <stdio.h>
2
3 int main() {
4     int arr[100], n, i, j, k;
5
6     scanf("%d", &n);
7
8     for (i = 0; i < n; i++) {
9         scanf("%d", &arr[i]);
10    }
11
12    for (i = 0; i < n; i++) {
13        for (j = i + 1; j < n; j++) {
14            if (arr[i] == arr[j]) {
15                for (k = j; k < n - 1; k++) {
16                    arr[k] = arr[k + 1];
17                }
18                n--;
19                j--;
20            }
21        }
22    }
```

OUTPUT

Array after removing duplicates:
1 2 3

5
1 1 2 3 3

QUESTIONS 14 / 25 completed

Kth Min

KTH MIN

Write a C program to find the kth smallest element in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, k, temp;
5
6     printf("Enter the size of the array: ");
7     scanf("%d", &n);
8
9     int arr[n];
10
11    printf("Enter the array elements: ");
12    for(i = 0; i < n; i++) {
13        scanf("%d", &arr[i]);
14    }
15
16    printf("Enter the value of k: ");
17    scanf("%d", &k);
18
19    // Sort the array
20    for(i = 0; i < n - 1; i++) {
21        for(j = i + 1; j < n; j++) {
```

OUTPUT

Enter the size of the array: Enter the array elements: Enter the value of k: The 2th smallest element is 2

5
1 5 7 2 3
2

QUESTIONS 14 / 25 completed

Missing

MISSING

Write a C program to find the missing number in an array of integers.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, sum = 0, total, missing;
5
6     scanf("%d", &n);
7
8     int arr[n - 1];
9
10
11     for(i = 0; i < n - 1; i++) {
12         scanf("%d", &arr[i]);
13         sum += arr[i];
14     }
15
16     total = n * (n + 1) / 2;
17     missing = total - sum;
18
19     printf("Missing number = %d\n", missing);
20
21     return 0;
}
```

OUTPUT

Missing number = 4

INPUT

5
1 2 3 5

QUESTIONS 14 / 25 completed

Unique

UNIQUE

Write a C program to print all the unique elements in an array.

PROGRAM EDITOR

C

Run

Save

```
1 #include <stdio.h>
2
3 int main() {
4     int n, i, j, count;
5
6
7     scanf("%d", &n);
8
9     int arr[n];
10
11
12     for(i = 0; i < n; i++) {
13         scanf("%d", &arr[i]);
14     }
15
16     printf("Unique elements are: ");
17
18     for(i = 0; i < n; i++) {
19         count = 0;
20
21         for(j = 0; j < n; j++) {
```

OUTPUT

Unique elements are: 1 2

INPUT

6
1 2 3 3 4 4